Gene-Query JEPA · normal → 90m / 6h / 10h

Source is a normal cell. Targets are $t \in \{1, 2, 3\}$: 90m, 6h, 10h, one pass each. Time enters only as $T[t]$.

ONLINE · frozen

Source cell

normal tokens x^{src}

$x^{\text{src}} \in \mathbb{N}^{B \times L_s}$, $L_s \le 512$

Encoder

MaskGIT, layers 1–3 of 12

time index fixed at 1 · θ frozen

H_{src}

$H_{\text{src}} \in \mathbb{R}^{B \times L_s \times 768}$

this cell only

memory

PREDICTOR · the only thing that trains

Query

$q = E[g] + R[r_{\text{src}}(g)] + T[t]$

$E[g],\ R[\cdot],\ T[t] \in \mathbb{R}^{768}$

$Q \in \mathbb{R}^{B \times 129 \times 768}$ (128 genes + 1 cell)

Predictor

3-layer decoder, 8 heads

cross-attends only to this cell's H_{src}

96 uniform from the source list, 32 from outside it

20% of the 96 drop $E[g]$ (mask-CE)

$\hat{z}$

$\hat{z}_{\text{gene}} \in \mathbb{R}^{B \times 128 \times 768}$

$\hat{z}_{\text{cell}} \in \mathbb{R}^{B \times 768}$

EMA TARGET · no gradient

Target cell

$t \in \{1, 2, 3\}$

90m, 6h, 10h · one pass each

EMA encoder

frozen copy of the online encoder

stop-gradient · $\bar{\theta}$ fixed

Answer $z^{\star}$

$z^{\star} \in \mathbb{R}^{B \times 128 \times 768}$

present → row of H_{tgt}

$H_{\text{tgt}} \in \mathbb{R}^{B \times L_t \times 768}$, $L_t \le 512$

absent → $a_{\text{null}} \in \mathbb{R}^{768}$

Loss

mean over $t \in \{1, 2, 3\}$

$\mathcal{L}_{\text{var}}$ $\lambda = 1$	$\mathcal{L}_{\text{abs}}$ $\lambda = 0.25$	$\mathcal{L}_{\text{NCE}}$ $\lambda = 0.1$	$\mathcal{L}_{\text{mask}}$ $\lambda = 0.1$
present genes raw squared error	missing genes pulled to the null	same gene, same cell against the batch	name the gene after hiding it